

2026-02-24 - HFR-protocol - Full.wep

#	Step Name	
1	HFR protocol initial settings	
2	PLAX 2D	Clinical part of echo study starts here.
3	PLAX 2D + colour	Color flow cine loop in PLAX
4	Parasternal bi-plane	
5	PLAX zoomed MV	
6	PLAX zoomed MV + colour	
7	PLAX zoomed AV i)	
8	Parasternal zoomed bi-plane of AV	
9	PLAX zoomed AV + colour	
10	PLAX tilted anterior (towards RV)	
11	PLAX tilted anterior+ colour	
12	PLAX tilted anterior+ colour + CW	
13	PLAX aortic level	
14	PSAX AV level	
15	PSAX zoomed to AV	
16	PSAX zoomed to AV + colour	
17	PSAX zoomed AV 3D	
18	PSAX TV at AV level + colour	
19	PSAX TV at AV level + CW	
20	PSAX PV at AV level	
21	PSAX PV at AV level + colour	
22	PSAX PV at AV level + CW	
23	PSAX PV at AV level + PW	
24	PSAX at MV level	
25	PSAX at MV level + colour	
26	PSAX at papillary muscles level	
27	PSAX at apical level	
28	A4C, whole heart	
29	A4C + colour across MV	Color flow cine loop in apical 4 chamber
30	Tri-plane LV and LA	
31	Tri-plane LV	
32	A4C 3D volume of LV single-beat	
33	A4C 3D volume of LV multi-beat	
34	A4C LV	
35	A4C LV + TDI	TDI for apical 4 chamber
36	A4C LV basal septum + TDI + PW	

37	A4C LV basal lateral + TDI + PW	
38	A4C LV basal lateral + M-mode	
39	A4C LV basal medial + M-mode	
40	A4C zoomed LA and MV	
41	A4C zoomed LA and MV + TDI	
42	A4C zoomed LA with PV inflow + color + PW	
43	A4C zoomed MV inflow + colour + PW	
44	A5C zoomed LVOT and AV	
45	A5C zoomed LVOT and AV + colour	
46	A5C zoomed LVOT and AV + colour + CW i)	
47	A5C zoomed LVOT and AV + colour + CW ii)	
48	A5C zoomed LVOT and AV + colour + PW i)	
49	A5C zoomed LVOT and AV + colour + PW ii)	
50	A5C zoomed to AV and MV level + PW	
51	A2C	
52	A2C + colour across MV	Color flow cine loop in apical 2 chamber
53	A2C zoomed to LV	
54	A2C zoomed to LV + TDI	TDI for apical 2 chamber
55	A2C + TDI + PW basal inferior	
56	A2C + TDI + PW basal anterior	
57	A2C M-mode basal anterior	
58	A2C M-mode basal inferior	
59	A2C zoomed LA and MV	
60	A2C zoomed LA and MV + TDI	
61	A2C zoomed to LA and MV + colour	
62	A3C	
63	A3C + colour across MV and AV	Color flow cine loop in apical 3 chamber
64	A3C zoomed to LV	
65	A3C zoomed to LV + TDI	TDI for apical 3 chamber
66	A3C + colour + PW LVOT	
67	A3C + CW	
68	A4C zoomed to septum	
69	A4C zoomed to septum + TDI	TDI for focused septum
70	A4C optimized for right heart	
71	A4C zoomed to RA and TV	
72	A4C zoomed to RA and TV + colour	

73	A4C zoomed to RA and TV + colour + CW	
74	A4C zoomed to RA and TV + PW	
75	A4C zoomed to RV i)	
76	A4C zoomed to RV ii)	
77	A4C zoomed to RV + TDI	
78	A4C zoomed to RV free wall + TDI + PW	
79	A4C zoomed to RV + M-mode	
80	Subc. 4C	
81	Subc. 4C zoomed to RV free wall	
82	VCI	
83	VCI zoomed bi-plane	
84	VCI + M-mode	
85	Suprasternal aorta desc.	
86	Suprasternal aorta desc. + colour	
87	Suprasternal aorta desc + colour + CW	
88	Suprasternal aorta desc + colour + PW	
89	Suprasternal notch - Pedoff	
90	Right parasternal - Pedoff	
91	Subcostal - Pedoff	
92	PLAX	
93	PLAX, HQ - image	HFR echo protocol starts here
94	PLAX, HFR plane wave 1	
95	PLAX, HFR plane wave 2 - data dump (I)	
96	PLAX, HFR plane wave 2 - data dump (II)	
97	PLAX, HFR diverging wave 1	
98	PLAX, HFR diverging wave 2 - data dump (I)	
99	PLAX, HFR diverging wave 2 - data dump (II)	
100	PLAX, TDI HQ	
101	PLAX, HFR TDI 1	
102	PLAX, HFR TDI 2	
103	PLAX, HFR TDI-IQ 1	
104	PLAX, HFR TDI-IQ 2	
105	A4C zoomed to interventricular septum - HQ	
106	A4C zoomed to interv. septum, HFR plane wave 1	
107	A4C zoomed to interv. septum, HFR plane wave 2 - data dump (I)	
108	A4C zoomed to interv. septum, HFR plane wave 2 - data dump (II)	

109	A4C zoomed to interv. septum, HFR diverging wave 1
110	A4C zoomed to interv. septum, HFR diverging wave 2 - data dump (I)
111	A4C zoomed to interv. septum, HFR diverging wave 2 - data dump (II)
112	A4C zoomed to interv. septum, TDI HQ
113	A4C zoomed to interv. septum, HFR TDI 1
114	A4C zoomed to interv. septum, HFR TDI 2
115	A4C zoomed to interv. septum, HFR TD-IQ 1
116	A4C zoomed to interv. septum, HFR TD-IQ 2
117	A4C - HQ
118	A4C, HFR plane wave 1
119	A4C, HFR plane wave 2 - data dump (I)
120	A4C, HFR plane wave 2 - data dump (II)
121	A4C, HFR plane wave 2 - data dump (III)
122	A4C, HFR diverging wave 1
123	A4C, HFR diverging wave 2 - data dump (I)
124	A4C, HFR diverging wave 2 - data dump (II)
125	A4C, HFR diverging wave 2 - data dump (III)
126	A4C, TDI - HQ
127	A4C, HFR TDI 1
128	A4C, HFR TDI 2
129	A4C, HFR TDI-IQ 1
130	A4C, HFR TDI-IQ 2
131	A2C - HQ
132	A2C, HFR plane wave 1
133	A2C, HFR plane wave 2 - data dump (I)
134	A2C, HFR plane wave 2 - data dump (II)
135	A2C, HFR plane wave 2 - data dump (III)
136	A2C, HFR diverging wave 1
137	A2C, HFR diverging wave 2 - data dump (I)
138	A2C, HFR diverging wave 2 - data dump (II)
139	A2C, HFR diverging wave 2 - data dump (III)
140	A2C, TDI - HQ
141	A2C, HFR TDI 1
142	A2C, HFR TDI 2
143	A2C, HFR TDI-IQ 1
144	A2C, HFR TDI-IQ 2

145	A3C - HQ	
146	A3C, HFR plane wave 1	
147	A3C, HFR plane wave 2 - data dump (I)	
148	A3C, HFR plane wave 2 - data dump (II)	
149	A3C, HFR plane wave 2 - data dump (III)	
150	A3C, HFR diverging wave 1	
151	A3C, HFR diverging wave 2 - data dump (I)	
152	A3C, HFR diverging wave 2 - data dump (II)	
153	A3C, HFR diverging wave 2 - data dump (III)	
154	A3C TDI - HQ	
155	A3C, HFR TDI 1	
156	A3C, HFR TDI 2	
157	A3C, HFR TDI-HQ 1	
158	A3C, HFR TDI-HQ 2	
159	A4C	3D HFR Optional - only in selected cases
160	Tri-plane LV without LA	
161	3D volume of LV multi-beat - HQ image 1	
162	3D volume of LV multi-beat - HQ image 2	
163	3D volume of LV multi-beat HFR 1	
164	A4C 3D volume of LV multi-beat HFR 2 - data dump	
165	A4C - HQ - image before BSI imaging	<u>BSI protocol starts</u>
166	A4C, colour across LV and MV - HQ	FR index 0
167	A4C, BSI imaging 1 (LV and LA) - between HQ and maximized frame rate (1)	intermediate step 1
168	A4C, BSI imaging 2 (LV and LA) - between HQ and maximized frame rate (2)	intermediate step 2
169	A4C, BSI imaging 3 (LV and LA) - maximized frame rate	max frame rate
170	A4C LV, colour (LV) - HQ	
171	A4C LV, BSI imaging 1 (LV) - between HQ and max FR (1)	
172	A4C LV, BSI imaging 2 (LV) - between HQ and max FR (2)	
173	A4C LV, BSI imaging 3 (LV) - max. FR	
174	A2C, HQ - image before BSI imaging	
175	A2C, colour across LV and MV - HQ	
176	A2C, BSI imaging 1 (LV and LA) - between HQ and max FR (1)	
177	A2C, BSI imaging 2 (LV and LA) - between HQ and max FR (2)	
178	A2C, BSI imaging 3 (LV and LA) - max FR	

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179	A2C, colour (LV) - HQ
180	A2C, BSI imaging 1 (LV) - between HQ and max FR
181	A2C, BSI imaging 2 (LV) - between HQ and max FR
182	A2C, BSI imaging 3 (LV) - max FR
183	A3C, HQ - image before BSI imaging
184	A3C, colour across LV, MV and AV - HQ
185	A3C, BSI imaging 1 (LV and LA) - between HQ and max FR (1)
186	A3C, BSI imaging 2 (LV and LA) - between HQ and max FR (2)
187	A3C, BSI imaging 3 (LV and LA) - max FR
188	A3C LV, colour (LV) -HQ
189	A3C LV, BSI imaging 1 (LV) - between HQ and max FR (1)
190	A3C LV, BSI imaging 2 (LV) - between HQ and max FR (1)
191	A3C LV, BSI imaging 3 (LV) - max FR
192	PLAX, HQ - image before BSI imaging
193	PLAX , colour across LV, MV and AV - HQ
194	PLAX - BSI imaging 1 (LV, LA and LVOT) - between HQ and max FR (1)
195	PLAX - BSI imaging 2 (LV, LA and LVOT) - between HQ and max FR (2)
196	PLAX - BSI imaging 3 (LV, LA and LVOT) - max FR